Introduction to Linux

For Data Scientist





KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN

fit@hcmus

Content

- Introduction to Linux
- Why use Linux in data science?
- Linux basic
- Demo
  - Basic commands
  - Commands for Data Science
- Bash Scripting

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Unix

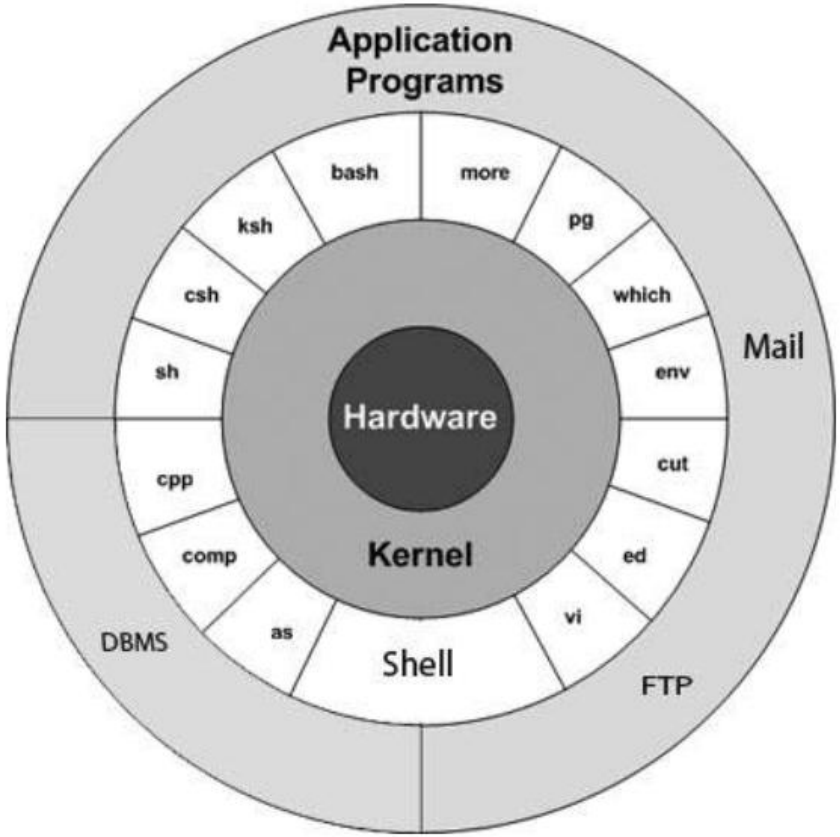
- Unix is a powerful, multiuser, multitasking operating system originally developed in 1969 by **Ken Thompson**, **Dennis Ritchie**, … at Bell Labs.
- It laid the foundation for modern operating systems and introduced concepts that are still fundamental today.

Unix

- **Mainly used for scientific-technical applications** on mainframes and workstations, but has become, because of Linux, also popular for classical PC-applications throughout the last years.
- Is perfectly suited for **application in networks**. Larger systems and networks require an administrator.
- Offers various alternatives for the solution of most tasks. The multitude of commands (more than in any other OS) are brief and flexible.
- Is originally **command-line oriented**, but can be used via a graphical user interface

Unix Architecture

- **Kernel:** The kernel is the heart of the OS. It interacts with the hardware and most of the important tasks.
- **Shell**: The shell is the utility that processes your requests. When you type in a command at your terminal, the shell interprets the command and calls the program that you want.
- **Commands and Utilities:** There are various commands and utilities which you can make use
- **Files and Directories:** all the data of Unix is organized into files. All files are then organized into directories.



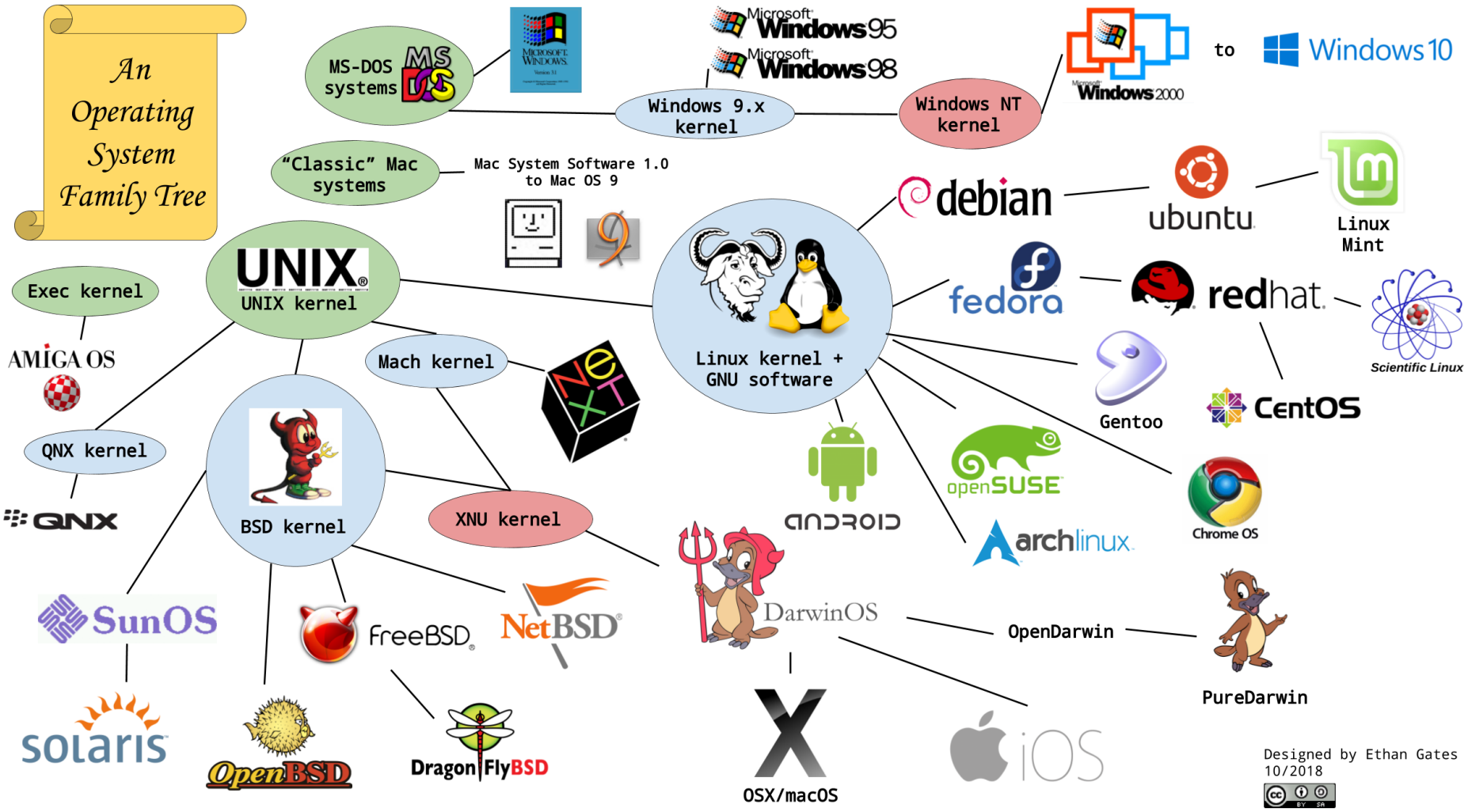
What is Linux?

- Linux is a free and open-source operating system **kernel** that, when combined with **GNU tools and utilities**, creates a complete operating system (often called GNU/Linux).
- It's **Unix-like** but not directly derived from Unix source code.

Linux

- **Linus** devised a ***kernel named Linux*** in 1991. (Lacking programs like File Manager, Document Editors, Audio-Video programs)
- He collaborated with other programmers in places like MIT and **applications** for Linux started to appear.
- The earlier versions of Linux OS were **not so user-friendly**
- This definitely curbed the Linux's popularity as other commercially oriented **Operating System Windows** got famous.
- Nonetheless, the **open-source aspect** of the Linux operating system made it more robust.

OPERATING SYSTEMS



https://github.com/EG-tech/digipres-posters

Linux

- The main advantage of Linux was that programmers were able to use the Linux Kernel to **design their own custom** operating systems.
- With time, a new range of **user-friendly OS**'s stormed the computer world.
- Now, Linux is one of the most popular and widely used Kernel



Linux in the Real World

- **Servers**: ~96% of the world's top 1 million web servers
- **Supercomputers**: 100% of the world's top 500 supercomputers
- **Mobile**: Android is built on Linux
- **Cloud**: AWS, Google Cloud, Azure all heavily use Linux
- **Data Science**: Most data science infrastructure runs on Linux

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Benefits of Linux

- Being **open-source**, anyone can modify it.
- It is **easy to learn** Linux for beginners
- The Linux operating systems now offer millions of programs/applications and Linux softwares to choose from, most of them are **free**!
- Once you have Linux installed you no longer need an antivirus! Linux is a **highly secure system**.
- Linux freeware is the OS of choice for Server environments due to its **stability and reliability** (Mega-companies like Amazon, Facebook, and Google use Linux for their Servers). A Linux based server could run non-stop without a reboot for years on end.

Why using Linux

- When working for companies in the future, it's likely that you have to interact with Linux computer, and probably have to interact with it through commands (e.g. when you use remote server)
- Or when you work with Google Colab, the only way to interact with Linux computer behind Google Colab is to use commands

Why using Linux

- **Industry Standard**: Most production servers run Linux
- **Open Source**: Free and customizable
- **Powerful CLI**: Efficient data manipulation
- **Package Management**: Easy software installation
- **Remote Access**: SSH for cloud computing
- **Scripting**: Automation capabilities
- **Performance**: Better resource management
- **Community**: Extensive documentation and support

What is Linux Distro

- A Linux distribution (distro) is a ***complete operating system*** that includes the ***Linux kernel***, *system libraries*, *utilities*, *application software*, and a *package management system*.
- It is created by assembling various software components from different sources and packaging them together to provide a cohesive and user-friendly computing environment.

What is Linux Distro

- Popular free Linux distributions:
  - Ubuntu from Canonical
  - Debian from the Debian Project
  - Fedora from the Fedora Project
  - OpenSUSE from SUSE
  - Rocky and AlmaLinux
- Commercial distributions:
  - Oracle Linux
  - Red Hat Enterprise Linux (RHEL)
  - SUSE Enterprise Server (SLES).
- ***Recommendation: Ubuntu 24.04 LTS or Ubuntu 26.04 LTS for beginners***

Linux vs. Windows

- Linux is an **open source operating system** so user can change source code as per requirement whereas Windows OS is a commercial operating system so user doesn’t have access to source code.
- Linux is very well **secure** as it is easy to detect bugs and fix whereas Windows has a huge user base, so it becomes a target of hackers to attack windows system.
- Comparing Windows file system vs Linux file system, Linux **runs faster** even with older hardware whereas Windows are slower compared to Linux.

Linux vs. Windows

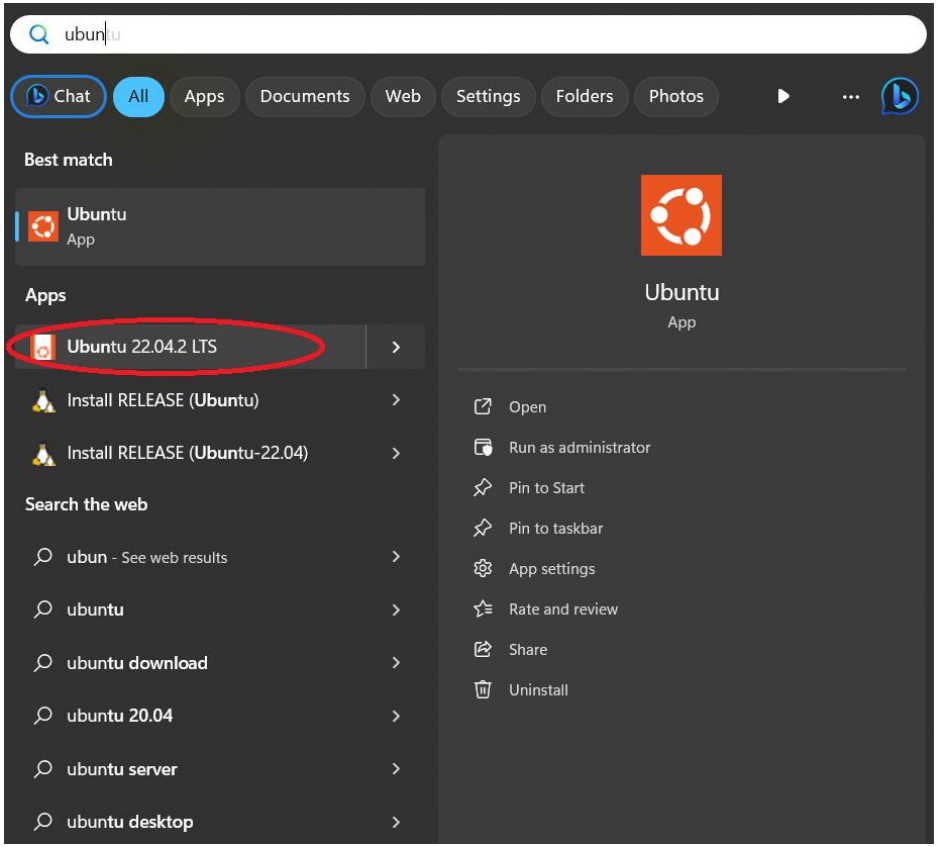
- Linux **peripherals** like hard drives, CD-ROMs, printers are considered files whereas Windows, hard drives, CD-ROMs, printers are considered as devices
- Linux files are ordered in a **tree structure** starting with the root directory whereas in Windows, files are stored in folders on different data drives like C: D: E:
- In Linux you would find the system and program files in different directories whereas in Windows, system and program files are usually saved in C: drive.
- Linux filename is **case-sensitive**. Windows is not

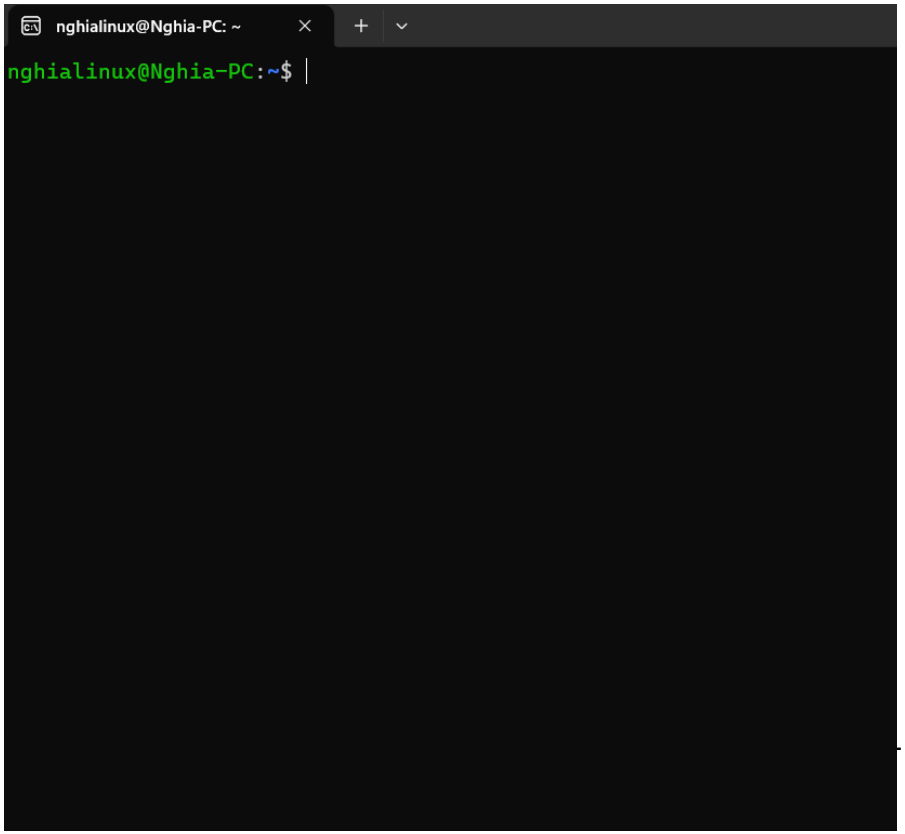
How to install Linux (on Windows)

- https://ubuntu.com/tutorials/install-ubuntu-on-wsl2-on-windows-11-with-gui-support
  - Install WSL
  - Download Ubuntu
  - Configure Ubuntu

How to install Linux (on Windows)

- After install:
  - Type: “Ubuntu”
  - Choose ubuntu version

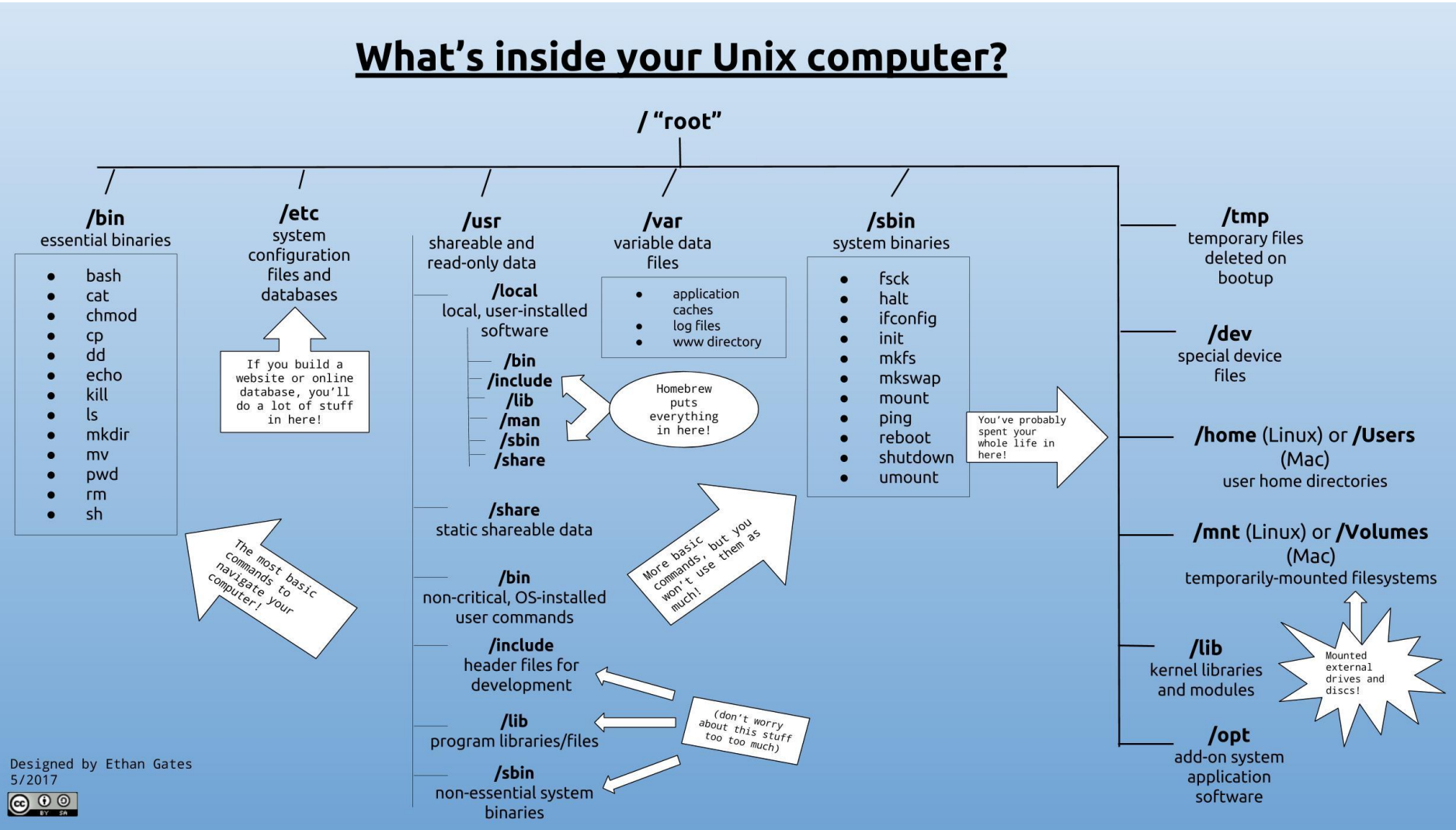




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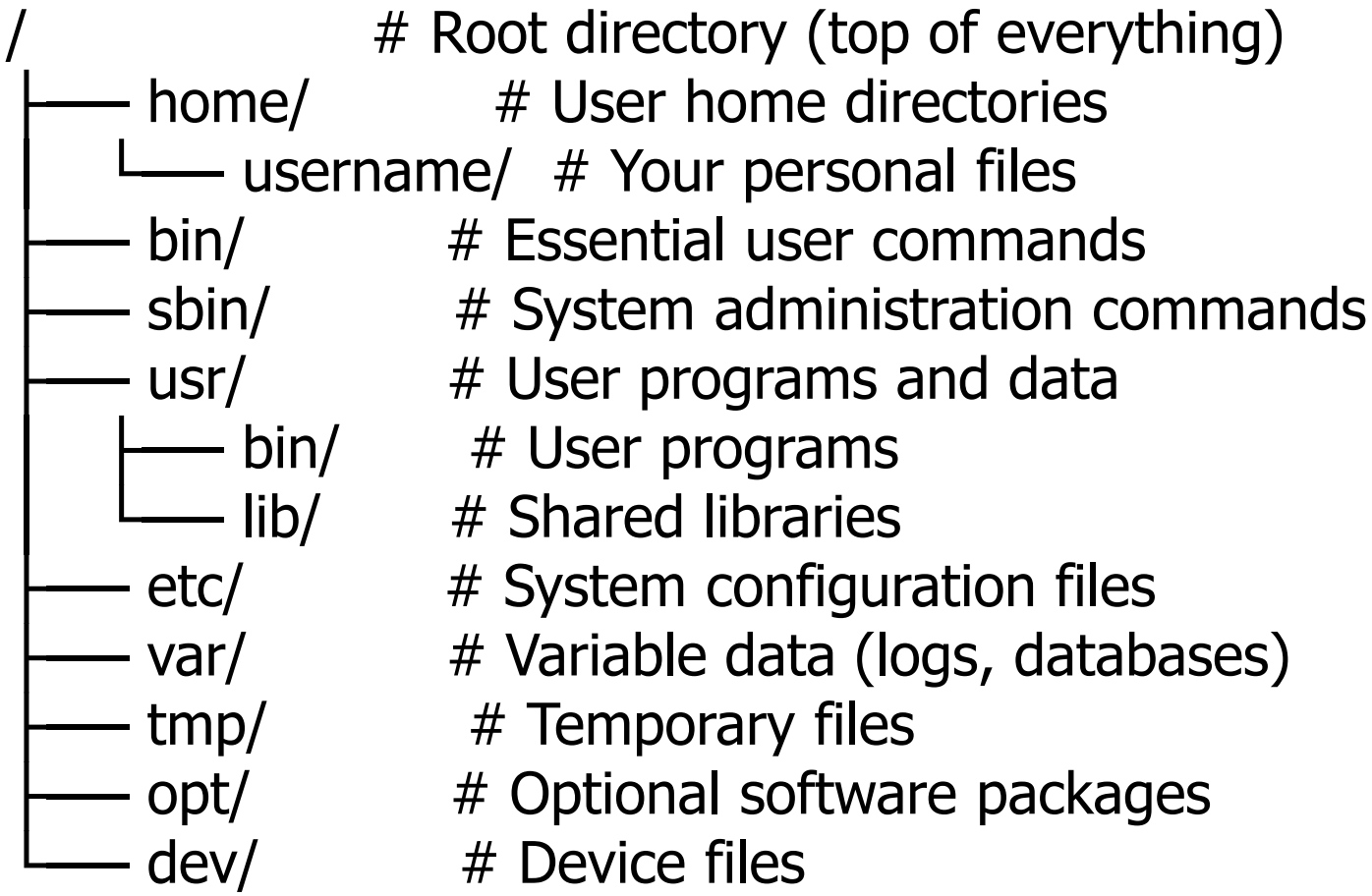
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Root



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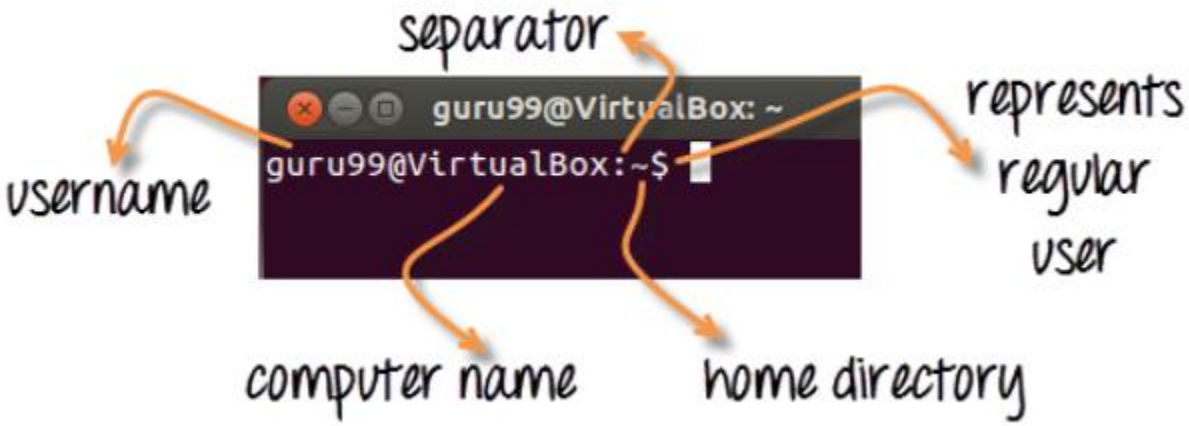
Important Directories



Command Line Interface

- Even though the world is moving to GUI based systems, CLI has its specific uses and is widely used in **scripting** and **server administration**.
  - Comparatively, Commands offer more options & are flexible. Piping and stdin/stdout are immensely powerful are not available in GUI
  - Some configurations in GUI are up to 5 screens deep while in a CLI it's just a single command
  - Moving, renaming 1000's of the file in GUI will be time-consuming while in CLI, using regular expressions so can do the same task with a single command.
  - CLI load fast and do not consume RAM compared to GUI. In crunch scenarios this matters.

Command Line Interface



Basic Command Structure

- command [options] [arguments]
- Example:

```
ls -la /home/user
|  |    └─ argument (what to operate on)
|  └─ options (how to operate)
└─ command (what to do)
```

Basic command line

- https://cheatography.com/davechild/cheat-sheets/linux-command-line/

Essential Navigation Commands

- Moving around

```
pwd                 # Print working directory
cd /path/to/dir     # Change directory
cd ~                # Go to home directory
cd ..               # Go up one level
cd -                # Go to previous directory
ls                  # List files
ls -la              # List all files with details
```

File and Directory Operations

- Basic Operations

```
mkdir myproject          # Create directory
mkdir -p data/raw/csv    # Create nested directories
touch file.txt           # Create empty file
cp source dest           # Copy file
cp -r dir1 dir2          # Copy directory
mv old new               # Move/rename
rm file.txt              # Remove file
rm -rf directory         # Remove directory recursively
```

Viewing File Contents

Display File Content

```
cat file.txt          # Display entire file
less file.txt         # Page through file
head -n 10 file.txt   # First 10 lines
tail -n 20 file.txt   # Last 20 lines
tail -f logfile.log   # Follow file updates (logs)
wc -l file.txt        # Count lines
```

Text Search and Filtering

Finding Content

```
grep "pattern" file.txt          # Search in file
grep -r "pattern" directory/     # Recursive search
grep -i "pattern" file.txt       # Case-insensitive
grep -v "pattern" file.txt       # Invert match
find . -name "*.csv"             # Find files by name
find . -type f -size +100M       # Find large files
```

File Permission

```
nghialinux@Nghia-PC:~/02-DemoDir$ ls -l
total 1044
drwxr-xr-x 4 nghialinux nghialinux    4096 Oct  1 21:37 dir1
-rw-r--r-- 1 nghialinux nghialinux      32 Sep 24  2021 file1.txt
-rw-r--r-- 1 nghialinux nghialinux      32 Sep 24  2021 file2.txt
-rw-r--r-- 1 nghialinux nghialinux 1053440 Sep 24  2021 long_file
```

- Owner permissions
- Group permissions
- Other (world) permissions
- Specific permission:
  - read (r)
  - write (w)
  - execute (x)
  - –

File Permission

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```

- First character indicates whether it is a normal file (-) or directory (d)
- Next 9 characters are permissions for the file or directory
- The next field is the number of blocks
- The next field is the owner of the file or directory
- The next field is the group the file or directory belongs to
- Following this is the file size.
- Next up is the file modification time.
- Finally we have the actual name of the file or directory.

File Permissions

- Changing Permissions

```
chmod 755 script.sh      # rwxr-xr-x
chmod +x script.sh       # Add execute permission
chown user:group file    # Change ownership
```

Pipes and Redirection

- Connecting Commands

```
# Output redirection
command > file.txt       # Overwrite file
command >> file.txt      # Append to file
command 2> error.log     # Redirect errors

# Pipes - connect commands
cat data.csv | grep "2024" | wc -l
ls -la | grep "csv"
history | grep "python"
```

Package Management

- Managing Packages

```
# Install pip packages
pip install pandas numpy scikit-learn

# Install from requirements
pip install -r requirements.txt

# List installed packages
pip list

# Show package info
pip show pandas

# Uninstall package
pip uninstall package-name
```

Managing Python Packages

- APT Package Manager

```
# Update package lists
sudo apt update

# Upgrade installed packages
sudo apt upgrade

# Install package
sudo apt install python3-pip

# Remove package
sudo apt remove package-name

# Search for package
apt search keyword
```

Virtual Environments

- Isolating Python Projects

```
# Create virtual environment
python3 -m venv myenv

# Activate environment
source myenv/bin/activate

# Deactivate
deactivate

# Using conda
conda create -n ds_env python=3.10
conda activate ds_env
conda install pandas numpy matplotlib
```

System Monitoring

- Check System Resources

```
# Disk usage
df -h                 # Disk space
du -sh directory/     # Directory size
du -h --max-depth=1   # Subdirectory sizes

# Memory usage
free -h

# CPU info
lscpu
cat /proc/cpuinfo

# GPU info (NVIDIA)
nvidia-smi
```

SSH and Remote Access

- Connecting to Remote Servers

```
# Basic SSH connection
ssh username@hostname

# SSH with key
ssh -i ~/.ssh/id_rsa username@hostname

# Copy files to remote (SCP)
scp file.txt username@hostname:/remote/path/

# Copy from remote
scp username@hostname:/remote/file.txt ./

# Recursive copy
scp -r directory/ username@hostname:/remote/path/
```

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Prepare

- [ ] 02-DemoDir.rar -> Extract
  - [ ] Put in home folder
- [ ] 02-DemoLinuxCommand.html
- [ ] Data/movies.csv

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Bash Scripting

- A Bash script: define a series of actions which the computer will then perform without us having to enter the commands ourselves.
  - If a particular task is done often, or it is repetitive, then a script can be a useful tool.
- A Bash script is interpreted
- Anything you can run on the command line you may place into a script and they will behave exactly the same.
  - Anything you can put into a script, you may run on the command line and again it will perform exactly the same.

Bash Scripting: Example

- Create a script file (DemoScript.sh)

```
#!/bin/sh
# Simple demo script
echo Here is the files in your directory:
ls -l
```

  - ***Beware if create file on windows***
- **First line**:
  - The very first line of a script should always be this line.
  - This line identifies which interpreter should be used.
  - The first two characters "`#!`" are referred to as a shebang. After that (important, no spaces) is the path to the interpreter.
- Linux is an extensionless system. ".sh" extension is is purely for convenience and is not required

Bash Scripting: Example

- Run the script

```
(base) nghialinux@Nghia-PC:~/02-DemoDir$ ls -l myscript.sh
-rw-r--r-- 1 nghialinux nghialinux 50 Oct 10 11:17 myscript.sh
(base) nghialinux@Nghia-PC:~/02-DemoDir$ chmod 755 myscript.sh
(base) nghialinux@Nghia-PC:~/02-DemoDir$ ls -l myscript.sh
-rwxr-xr-x 1 nghialinux nghialinux 50 Oct 10 11:17 myscript.sh
(base) nghialinux@Nghia-PC:~/02-DemoDir$ ./myscript.sh
Here are....
dir1  file1.txt  file2.txt  long_file  myscript.sh
(base) nghialinux@Nghia-PC:~/02-DemoDir$
```

- chmod 755 myscript.sh: set execute right.
- **./**myscript.sh: run the script
  - ./myscript.sh we are actually tellling the system to look in our current directory to find the script

Bash Scripting: Variables

- Create a script file (variableexample.sh)

```
#!/bin/bash
# A simple demonstration of variables

name='Ryan'
echo Hello $name
```

- When we set a variable, we specify it's name, followed directly by an equals sign (=) followed directly by the value.
  - No spaces on either side of the = sign.
- When we refer to a variable, we must place a dollar sign ($) before the variable name.

Bash Scripting: Command line arguments

- Create a script file (commandlineagvDemo.sh)

```
#!/bin/sh
# A simple demonstration of variables

echo My name is $0 and I have been given $# command line arguments
echo Here they are: $*
echo And the 2nd command line argument is $2
```

- $0 - The name of the script.
- $1 - $9 - Any command line arguments given to the script. $1 is the first argument, $2 the second and so on.
- $# - How many command line arguments were given to the script.
- $* - All of the command line arguments.

A Sample Backup Script

- Create a script file (projectbackup)

```
#!/bin/bash
# Backs up a single project directory

date=`date +%F`
mkdir backups/$1_$date
cp -R $1 backups/$1_$date
echo Backup of $1 completed
```

- *For more info about bash scripting (if statement, loop, function…) can check this Bash Scripting Tutorial*

Simple Tutorial

- https://missing.csail.mit.edu/2020/course-shell/
- https://ubuntu.com/tutorials/command-line-for-beginners
- https://missing.csail.mit.edu/
- https://ryanstutorials.net/linuxtutorial/
- The Linux Command Line (book by William Shotts)

Reference

- [1] The operating system Linux and programming languages: An introduction, Joachim Puls and Michael Wegner
- [2] https://www.tutorialspoint.com/unix/
- [3] https://www.guru99.com/introduction-linux.html